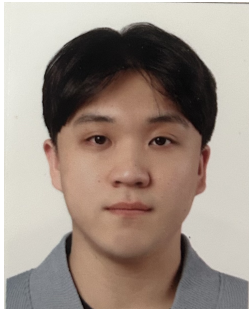


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SUMMARY

My belief is simple: pursue a problem until it breaks, and prove what others dismiss as impossible. That drive took me from a history degree to self-directed software training through the 42 curriculum, where I learned to work across the full systems stack — partitioning automotive workloads with the Xen hypervisor, building custom Linux images with Yocto, containerizing services with Docker and Kubernetes, and deploying ML models that solve real-world logistics problems. Wherever the layer — kernel, container, or model — I want to build systems that are understood down to their foundations and proven on real hardware and real data.

EDUCATION

Korea National Open University – PRIME College <i>B.S. in Artificial Intelligence</i>	Mar 2026 – Present
42 Wolfsburg – SEA:ME Program <i>Automotive Embedded AI · Sponsored by Volkswagen Korea</i>	Jan 2026 – Present
42 Gyeongsan Common Core <i>Software engineering curriculum · C, C++, Unix, algorithms, systems programming</i>	Jan 2024 – Jan 2026
Yeungnam University <i>B.A. in History</i>	Mar 2018 – Feb 2024

EXPERIENCE

Samseonghyeon History and Culture Museum · Researcher <i>Gyeongsan, South Korea</i>	Mar 2025 – Dec 2025
- Managed and analyzed artifact datasets; organized and maintained collection databases for historical records and exhibition documentation - Maintained and administered museum software systems and internal digital infrastructure - Prepared public-facing written materials, including exhibit text and social media content	

TECHNICAL SKILLS

Languages: C, C++, Python, Shell Script, OCaml
Embedded: Yocto / BitBake, Xen Hypervisor, RTOS, CAN Bus, Raspberry Pi 4
Software: Linux, Git, Docker, Docker Compose, Kubernetes (K3s / K3d), Argo CD, Vagrant, NGINX, Qt / QML, OpenGL
AI / ML: PyTorch, TensorFlow, LightGBM, Pandas, NumPy, Scikit-learn, Time-Series Forecasting

PROJECTS

Integrated Digital Cockpit · SEA:ME Capstone <i>Xen Hypervisor, Yocto / BitBake, Raspberry Pi 4, Qt / QML, CAN Bus</i>	2026
- Built a mixed-criticality automotive cockpit system on a Raspberry Pi 4 using Xen hypervisor-based Dom0 / DomU separation - Integrated a digital cluster, head unit Qt application, and parking distance control system into a custom Yocto-based Linux image - Ported the full stack onto an RC car platform and validated live hardware operation	
Autobahn Truck Parking Availability Prediction · Applied Deep Learning <i>Python, PyTorch, LightGBM, Pandas, NumPy, Kubernetes, TimescaleDB, Feast, Redis</i>	2026 – Present
Developing a route-aware deep learning model to forecast truck parking availability at German Autobahn rest areas, helping drivers estimate legal and safe stops under driving-time and rest-period regulations	

- Ingesting and processing German Toll Collect SID traffic data to extract spatial and temporal demand patterns across motorway parking sites
- Building feature pipelines covering time, location, route context, and parking demand; trained a LightGBM baseline achieving production-grade accuracy
- Deployed the full inference stack on Kubernetes with TimescaleDB for time-series storage, Feast for feature serving, and Redis for low-latency prediction caching

ContainerWeb — Containerized Web Infrastructure · System Administration

2026

Docker, Docker Compose, NGINX, MariaDB, WordPress / PHP-FPM, Bash

- Built a multi-container LEMP stack (NGINX, WordPress + PHP-FPM, MariaDB) from custom Debian images, orchestrated declaratively with Docker Compose
- Configured NGINX as a TLS-terminating reverse proxy (TLSv1.2 / 1.3, self-signed certificates) routing PHP requests to PHP-FPM over FastCGI
- Hardened the stack with Docker secrets, an isolated bridge network, and bind-mount volumes for persistence; automated build and lifecycle via a Makefile and idempotent shell scripts

Inception-of-Things (IoT) — Kubernetes GitOps · System Administration

2026

K3s, K3d, Vagrant, Argo CD, Kubernetes Ingress, Docker

- Provisioned reproducible multi-node K3s clusters with Vagrant (server / agent roles) and exposed workloads through host-based Ingress routing
- Built a K3d-based local Kubernetes environment and implemented a GitOps continuous-delivery pipeline with Argo CD syncing manifests from a Git repository
- Automated version rollouts by updating manifests in Git and verifying Argo CD auto-sync across successive application versions

AWARDS

Grand Prize · ESG Living Lab Matching Hackathon

Dec 2024

Ministry of Education, Republic of Korea

- Led a team project in partnership with advertising agency 'Terms One', driving the concept through to an award-winning final presentation
- Architected the overall structure and workflow for a bespoke ESG-focused internal software system
- Aligned the final proposal with partner-company specifications through strategic design discussions

Rising Star Award · 2025 Hackathon by 42 in Asia

Nov 2025

CDG · 42 Gyeongsan

- Built an Intelligent Document Processing (IDP) system to digitize complex multilingual government documents
- Engineered a modular AI agent pipeline using PaddleOCR-VL (OCR), XLM-RoBERTa (classification), and Ollama (summarization)
- Containerized the full system in Docker for rapid deployment